A Same patient, different policies

Held-out diagnostics case with hidden high (emergency) risk. Same simulator, same verifier.

Exact adaptive oracle

reference procedure: 4 questions

Patient sim

Adult with acute abdominal pain. No results on file yet.

Agent

Is the abdomen rigid?

Yes, the abdomen is rigid.

Is there rebound tenderness?

Yes, rebound tenderness is present.

Does CT show free air?

Yes, CT shows free air.

Is lactate elevated?

Yes, lactate is elevated.

Decision

Emergency treatment

Hidden risk: high
Decision: emergency treatment
Reward +0.83

Outcome-only GRPO

modal rollout (6 of 12 samples): 1 question

Patient sim

Adult with acute abdominal pain. No results on file yet.

Agent

Is the abdomen rigid?

Yes, the abdomen is rigid.

Decision

premature commitment **Admit**

Never asked: rebound tenderness, CT free air, lactate.

Hidden risk: high
Decision: admit
Reward +0.17

B Oracle procedures in the three tasks

One held-out case per task, exact adaptive oracle trajectory (Appendix B).

Diagnostics

acute chest pain

Patient sim

Adult with acute chest pain. No results on file yet.

Agent

Is troponin elevated?

Yes, troponin is elevated.

Does the ECG show ischemia?

Yes, the ECG shows ischemia.

Is the dyspnea severe?

Yes, the dyspnea is severe.

Is the patient hypotensive?

Yes, the patient is hypotensive.

Decision

Emergency treatment

Hidden risk: high
Decision: emergency treatment
Reward +0.83

Refill triage

lisinopril, no recent labs

Patient sim

Refill request: lisinopril (ACE inhibitor). No recent labs on file.

Agent

Is the eGFR declining?

Yes, the eGFR is declining.

Is the eGFR decline severe?

No, the decline is mild.

Is the patient dehydrated?

Yes, the patient is dehydrated.

Decision

Adjust dose

Hidden risk: moderate
Decision: adjust dose
Reward +0.81

Titration

older adult with atrial fibrillation

Patient sim

Older adult with atrial fibrillation. No risk factors documented yet.

Agent

Any history of major bleeding?

No, there is no prior bleeding.

Is the patient at high fall risk?

No, fall risk is low.

Decision

Full-intensity anticoagulation

Hidden risk: high clot, low bleed
Decision: full intensity
Reward +1.34